Lecture 12

Partitions, branching and Young's orthogonal form

Objectives. *Lecture 11 proved that every weight of S_n is a content vector, the content vector of a standard Young tableau. This lecture proves the converse by counting: admissible transpositions link exactly the tableaux of one shape, and Lectures 2 and 7 give $p(n)$ irreducibles. So partitions label the irreducibles, tableaux label their Gelfand–Tsetlin bases, and restriction removes one box. A coherent choice of phases then gives the matrices of the s_i: Young's orthogonal form.*

By the end of this lecture you will be able to:

- Decide from the axial distance whether exchanging i and $i+1$ in a standard tableau gives another standard tableau, and link any two standard tableaux of one shape by admissible transpositions.
- Prove that the irreducibles of S_n are the V^λ, $\lambda \vdash n$, with dimension the number of standard tableaux of shape λ and, for $n \geq 2$, $\operatorname{Res}^{S_n}_{S_{n-1}} V^\lambda \cong \bigoplus_{\mu=\lambda-\square} V^\mu$.
- Prove Young's orthogonal form, and explain why the phases of an orthonormal GZ basis must be fixed together.
- Write the matrices of $s_1, \dots, s_{n-1}$ on a given V^λ, and compute character values from them.

12.1 Shapes and admissible transpositions

For $T \in \mathrm{SYT}(n)$, write

$$\alpha(T) = (c_1(T), \dots, c_n(T)).$$

By Proposition 11.14, every content vector of length n is $\alpha(T)$ for exactly one $T \in \mathrm{SYT}(n)$, and its *shape* is the shape of T.

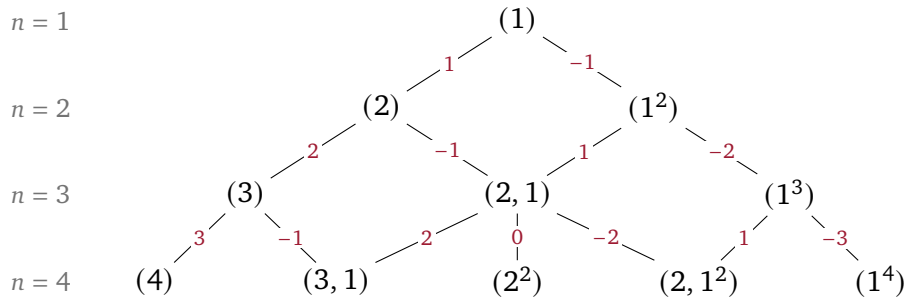
Definition 12.1 (Young's graph). *Young's graph* $\mathbb{Y}$ has as vertices the non-empty partitions, and an edge $\mu \to \lambda$ when the diagram of λ is that of μ with one box added: $\lambda = \mu + \square$ and $\mu = \lambda - \square$. The edge is labelled by the content of the added box.

A tableau $T \in \mathrm{SYT}(\lambda)$ is a path in $\mathbb{Y}$. Entries increase towards the lower right (Definition 11.13), so the boxes holding $1, \dots, k$ form a diagram, of shape $\lambda^{(k)}$ say, and

$$T \longmapsto \left(\lambda^{(1)} \to \lambda^{(2)} \to \cdots \to \lambda^{(n)} = \lambda\right).$$

The path starts at the box of 1, and step $k \geq 2$ adds the box of k, with label $c_k(T)$. Conversely, writing 1 in $(1,1)$ and k in the box added at step k turns a path from (1) to λ into a standard tableau, as each box comes after those above it and to its left. So $\mathrm{SYT}(\lambda)$ is the set of these paths, and the labels along the path of T are $\alpha(T)$ without its first entry, 0. In the graph

below, row n lists the partitions of n; each edge joins a partition to one obtained by adding a box, and its label is the content of that box. The graph up to $n = 4$ is



This is Example 10.11's branching graph of S_4, with the same labels. Theorem 12.6 explains why.

Definition 12.2 (Axial distance)**.** For $T \in \mathrm{SYT}(n)$ and $1 \le i \le n-1$, the *axial distance* is $r = c_{i+1}(T) - c_i(T)$: Lecture 11's r for the vector $\alpha(T)$. It is the change in column number minus the change in row number.

For a tableau T, let s_iT be the filling obtained by exchanging the entries i and $i+1$. This filling need not be standard.

Lemma 12.3 (Admissible transpositions of tableaux)**.** *Let $T \in \mathrm{SYT}(n)$, $1 \le i \le n-1$, and r the axial distance.*

(i) If i and $i+1$ lie in one row of T, then $r = 1$; if they lie in one column, then $r = -1$. In both cases s_iT is not standard.

(ii) Otherwise s_iT is a standard tableau of the shape of T, with

$$\alpha(s_iT) = s_i\alpha(T),$$

and $|r| \ge 2$.

So s_i is admissible for $\alpha(T)$ if and only if s_iT is standard.

Proof. (i) A box between i and $i+1$ in their row would hold an entry strictly between them. So $i+1$ is the right-hand neighbour of i, and $r = 1$; in a column, $i+1$ lies just below i, and $r = -1$. In s_iT, $i+1$ then precedes i in that row or column.

(ii) *Standard.* Any third entry is either less than both $i, i+1$ or greater than both, so exchanging them changes only their comparison with each other. They share no row or column, and s_iT is standard. The box of i in s_iT is the box of $i+1$ in T, and conversely, so $\alpha(s_iT) = s_i\alpha(T)$.

$r \ne 1$. Let i lie in (p, q) and $i+1$ in (p', q'), with $p' \ne p$ and $q' \ne q$. If $r = 1$, then

$$q' - p' = q - p + 1.$$

If $p' < p$, then $q' \le q$, and $q' \ne q$ gives $q' < q$. So (p', q') lies above and to the left of (p, q), and its entry $i+1$ is smaller than i, a contradiction. If $p' > p$, then $q' > q+1$. Since rows do not lengthen downwards, $\lambda_p \ge \lambda_{p'} \ge q' > q+1$, so $(p, q+1)$ is a box. It lies to the right of (p, q) and above and to the left of (p', q'). Its entry is therefore greater than i and less than $i+1$ (Definition 11.13), again impossible.

$r \ne -1$. The same argument applies with rows and columns exchanged.

$r \neq 0$. By the content-vector condition (C3) for standard tableaux (Proposition 11.14), equal entries of $\alpha(T)$ have other entries between them. Consecutive entries cannot be equal. Since r is an integer and is not 0 or ± 1, it has $|r| \geq 2$. ∎

12.2 Connectivity and the classification

Two content vectors are *linked* if a chain of transpositions, each admissible for the vector it is applied to, takes one to the other. As s_i is an involution, admissible for α if and only if for $s_i\alpha$, this is an equivalence relation.

Definition 12.4 (Row tableau)**.** For $\lambda \vdash n$, the *row tableau* T^λ fills its first row with $1, \dots, \lambda_1$, its second with the next λ_2 numbers, and so on. It is standard: each row increases, and entries in a lower row exceed those in every higher row.

Theorem 12.5 (Shapes are linked classes)**.** *Two content vectors of length n are linked if and only if they have the same shape.*

Proof. *Only if.* By Lemma 12.3(ii), an admissible s_i takes $\alpha(T)$ to $\alpha(s_iT)$ of the same shape. Applying this along a linking chain shows that its endpoints have the same shape.

If. By Lemma 12.3, it suffices to link each $T \in \mathrm{SYT}(\lambda)$ to the row tableau T^λ (Definition 12.4) by moves $T \mapsto s_iT$ with s_iT standard. We argue by induction on n; the case $n = 1$ is clear.

Moving n. Let b be the last box of the last row of λ, which holds n in T^λ, and k the entry of T in b. Apply $s_k, s_{k+1}, \dots, s_{n-1}$ in turn; if $k = n$, there are no moves.

Before s_j is applied, j is in b. Then $j+1$ lies neither in the row of b, which ends at b, nor in its column: nothing lies below b, and the entries above it are smaller than j. So s_j moves $j+1$ into b and keeps the tableau standard (Lemma 12.3(ii)).

The rest. Now n is in b, and the other boxes form a standard tableau of shape $\lambda - b$. By induction it is linked to $T^{\lambda-b}$ by moves s_i with $i \leq n-2$, which fix n. Each move keeps the full tableau standard, since n remains in the corner b; its admissibility depends on the same two contents as in the smaller tableau. And $T^{\lambda-b}$ with n in b is T^λ. ∎

For a partition λ, let $f^\lambda = |\mathrm{SYT}(\lambda)|$ be the number of standard tableaux of shape λ.

Theorem 12.6 (The classification)**.** *Let $n \geq 1$.*

(i) $\mathrm{Spec}(n) = \mathrm{Cont}(n)$.

(ii) For each $\lambda \vdash n$ exactly one $V^\lambda \in \mathrm{Irr}(S_n)$ has as its weights the content vectors of shape λ, and $\lambda \mapsto V^\lambda$ is a bijection from the partitions of n onto $\mathrm{Irr}(S_n)$.

(iii) V^λ has a GZ basis $\{v_T : T \in \mathrm{SYT}(\lambda)\}$ with $X_k v_T = c_k(T)\, v_T$; so $\dim V^\lambda = f^\lambda$.

(iv) If $n \geq 2$, then

$$\mathrm{Res}^{S_n}_{S_{n-1}} V^\lambda \cong \bigoplus_{\mu = \lambda - \square} V^\mu,$$

where the sum runs over the partitions obtained by removing one box.

Proof. *The weights of one irreducible.* Let $V \in \mathrm{Irr}(S_n)$. Its weights are content vectors (Theorem 11.12), and with a weight α they contain $s_i\alpha$ for every admissible s_i (Theorem 11.2(ii), as noted after Definition 11.5). By Theorem 12.5, if they contain one content vector of shape λ, they contain all of them. Different irreducibles have no weight in common (Corollary 10.15).

Counting. So we may choose for each V a shape λ_V whose content vectors are weights of V. Every shape has a tableau, for instance its row tableau, so two irreducibles assigned the same shape share a weight. By separation they are equal: $V \mapsto \lambda_V$ is one-to-one. By Theorem 7.11(ii) and Theorem 2.16,

$$\#\operatorname{Irr}(S_n) = \#\text{classes of } S_n = p(n).$$

So $V \mapsto \lambda_V$ is a bijection onto the partitions of n.

(i) and (ii). Every content vector is therefore a weight, which is (i). And V has no weight of a shape $\mu \neq \lambda_V$: those are weights of the V' with $\lambda_{V'} = \mu$. This is (ii), with V^λ the V with $\lambda_V = \lambda$.

The basis. By Corollary 10.15, $\gamma \mapsto \alpha(v_\gamma)$ maps the paths to V^λ one-to-one onto its weights, the $\alpha(T)$, $T \in \operatorname{SYT}(\lambda)$, one for each T (Proposition 11.14). Write v_T for the v_γ of weight $\alpha(T)$, and V_T for its line. These vectors are chosen up to non-zero scalars; § 12.3 fixes them coherently.

Branching: the last edge. Let $n \geq 2$, and let $\gamma = (U_1 \to \cdots \to U_n)$ be the path of v_T. By Proposition 10.10, a_k depends only on the edge $U_{k-1} \to U_k$, so the path $(U_1 \to \cdots \to U_{n-1})$ to U_{n-1} has the weight $(c_1(T), \dots, c_{n-1}(T))$. This is $\alpha(T')$, where T' is T with the box $\square$ of n removed, a standard tableau of shape $\lambda - \square$. By (ii) for S_{n-1}, $U_{n-1} = V^{\lambda-\square}$.

Branching: every box. If $\lambda - \square$ is a partition, some T holds n in $\square$: fill the rest by any standard tableau. So the edges into V^λ come from exactly the V^μ, $\mu = \lambda - \square$. By Maschke's theorem (Theorem 5.11(ii)) and multiplicity-freeness (Theorem 10.3), the restriction is their direct sum, each once. This is (iv). ∎

So the branching graph of Definition 10.4 is Young's graph, with V^λ at λ, and by the branching step, repeated, the path of v_T is the path of T. Each X_k is a sum of $k-1$ transpositions, which act by 1 on $\mathbf{1}$ and by -1 on sgn. These one-dimensional representations therefore have the weights of the one-row and one-column tableaux, respectively:

$$\mathbf{1} = V^{(n)}, \qquad \operatorname{sgn} = V^{(1^n)}.$$

For S_3, the plane E is $V^{(2,1)}$: its two weights are the content vectors of that shape (§ 11.5).

12.3 The orthogonal form

Fix $\lambda \vdash n$ and an invariant inner product on V^λ (Theorem 5.11(iii)). The lines V_T are orthogonal (Lemma 11.1(ii)), so V^λ has orthonormal GZ bases. Each line initially permits an independent phase, and relative phases affect the off-diagonal coefficients of s_i. We construct vectors with the positive coefficients of Young's form; then choosing one phase fixes all the others.

Definition 12.7 (σ_T and inversions)**.** For $T \in \operatorname{SYT}(\lambda)$, $\sigma_T \in S_n$ sends the entry of the row tableau T^λ (Definition 12.4) in each box to that of T. An *inversion* of T is a pair $j < k$ with k in a higher row of T than j; $\operatorname{inv}(T)$ is their number.

Lemma 12.8 (Moves and inversions)**.** *Let $T \in \operatorname{SYT}(\lambda)$.*

(i) If s_iT is standard, then $\sigma_{s_iT} = s_i\sigma_T$, and $\operatorname{inv}(s_iT)$ is $\operatorname{inv}(T)+1$ if $i+1$ lies in a lower row of T than i, and $\operatorname{inv}(T)-1$ if in a higher one.

(ii) $\operatorname{inv}(T^\lambda) = 0$. If $T \neq T^\lambda$, then for some i, s_iT is standard and $\operatorname{inv}(s_iT) = \operatorname{inv}(T) - 1$.

Proof. (i) *The permutation.* In each box, the entry of s_iT is s_i of that of T, so $s_i\sigma_T$ of that of T^λ.

The inversions. By Lemma 12.3, i and $i+1$ lie in different rows. For $j \neq i, i+1$, $j < i$ if and only if $j < i+1$, so the pairs $\{j,i\}$ and $\{j,i+1\}$ exchange roles. Only the pair $\{i,i+1\}$ changes: it is an inversion of s_iT exactly when $i+1$ lies lower than i in T.

(ii) In T^λ, every entry in a higher row is smaller than every entry in a lower row, so there are no inversions. If no i has $i+1$ in a higher row than i, the row indices of $1, 2, \dots, n$ weakly increase, forcing $T = T^\lambda$. Otherwise choose i with $i+1$ in a higher row than i. They are not in the same row, and cannot be in the same column: there $i+1$ would lie above the smaller entry i, against increase down the column. So s_iT is standard (Lemma 12.3), and the inversion-count formula applies. ∎

Theorem 12.9 (Young's orthogonal form). *Let u_{T^λ} be a GZ vector in V_{T^λ}, and for $T \in \mathrm{SYT}(\lambda)$ let u_T be the component of $\sigma_T u_{T^\lambda}$ in V_T. Then $u_T \neq 0$, and $v_T = u_T/\|u_T\|$ defines an orthonormal GZ basis of V^λ, a particular choice of the vectors v_T above, such that:*

(i) For $T \in \mathrm{SYT}(\lambda)$ and $1 \le i \le n-1$, with r the axial distance,

$$s_i v_T = r^{-1} v_T + \sqrt{1 - r^{-2}}\, v_{s_iT},$$

the second term absent when s_iT is not standard, where $r = \pm 1$.

(ii) An orthonormal GZ basis $\{v'_T\}$ of V^λ with $v'_T \in V_T$ satisfies (i) if and only if $v'_T = \theta v_T$ for every T, for one $\theta \in \mathbb{C}$ with $|\theta| = 1$.

Proof. *The vectors u_T.* By induction on $\mathrm{inv}(T)$: $u_T \neq 0$; $\sigma_T u_{T^\lambda} - u_T$ lies in the lines V_S with $\mathrm{inv}(S) < \mathrm{inv}(T)$; and whenever $T_1 = s_iT$ is standard with $\mathrm{inv}(T_1) = \mathrm{inv}(T) - 1$ and axial distance r_1,

$$\begin{aligned} u_T &= s_i u_{T_1} - r_1^{-1} u_{T_1}, \\ \|u_T\|^2 &= (1 - r_1^{-2}) \left\|u_{T_1}\right\|^2. \end{aligned} \tag{$*$}$$

If $\mathrm{inv}(T) = 0$, then $T = T^\lambda$ (Lemma 12.8(ii)) and $\sigma_T = e$.

The step. Let $\mathrm{inv}(T) = m \ge 1$, and $T_1 = s_iT$ any tableau as in $(*)$ (one exists, Lemma 12.8(ii)). By induction, $\sigma_{T_1} u_{T^\lambda} = u_{T_1} + x$, with $u_{T_1} \neq 0$ and x in the lines V_S with $\mathrm{inv}(S) < m-1$. As $\sigma_T = s_i\sigma_{T_1}$ (Lemma 12.8(i)),

$$\sigma_T u_{T^\lambda} = s_i u_{T_1} + s_i x.$$

The first term. As s_iT_1 is standard, $r_1 \neq \pm 1$; by Theorem 11.2(ii), with $w \in V_T$,

$$\begin{aligned} s_i u_{T_1} &= r_1^{-1} u_{T_1} + w, \\ \|w\|^2 &= (1 - r_1^{-2}) \left\|u_{T_1}\right\|^2. \end{aligned}$$

The rest. By Theorem 11.2 and Lemma 12.3, s_i maps each V_S into $V_S + V_{s_iS}$, or into V_S if s_iS is not standard, and $\mathrm{inv}(s_iS) < m$ when $\mathrm{inv}(S) < m-1$. So $u_T = w$, and the rest lies in lines V_S with $\mathrm{inv}(S) < m$.

(i) *Not standard.* If s_iT is not standard, then $r = \pm 1$, and $s_i v_T = r v_T$, which is $r^{-1} v_T$ (Lemma 12.3, Theorem 11.2(i)).

Upward. If $\mathrm{inv}(s_iT) = \mathrm{inv}(T) + 1$, then $(*)$ for s_iT, with $T_1 = T$, gives

$$\begin{aligned} s_i u_T &= r^{-1} u_T + u_{s_iT}, \\ \left\|u_{s_iT}\right\| &= \sqrt{1 - r^{-2}}\, \|u_T\|; \end{aligned}$$

divide by $\|u_T\|$.

Downward. If $\mathrm{inv}(s_iT) = \mathrm{inv}(T) - 1$, then $T_1 = s_iT$ has axial distance $-r$, and by $(*)$ and Theorem 11.2(ii), with $v = u_{T_1}$ and $w = u_T$,

$$s_i u_T = (1 - r^{-2})\, u_{T_1} + r^{-1} u_T;$$

divide by $\|u_T\| = \sqrt{1 - r^{-2}}\, \|u_{T_1}\|$.

(ii) Each v'_T is $\theta_T v_T$ with $|\theta_T| = 1$, and by (i) the coefficient of v'_{s_iT} in $s_i v'_T$ is

$$\theta_T \theta_{s_iT}^{-1} \sqrt{1 - r^{-2}}.$$

So (i) holds for $\{v'_T\}$ if and only if $\theta_{s_iT} = \theta_T$ whenever s_iT is standard; and any two tableaux of shape λ are joined by such moves (Theorem 12.5, Lemma 12.3). ∎

Example 12.10 (E in three bases; two guards)**.** For $\lambda = (2,1)$, the tableau $T = s_2 T^\lambda$ has $\sigma_T = s_2$ and $\mathrm{inv}(T) = 1$. In E take for u_{T^λ} the vector $u = (1, 1, -2)$, of weight $(0, 1, -1)$, and recall $f_1 = (1, -1, 0)$. The component of $s_2 u$ in the other GZ line is $\frac{3}{2} f_1$ (Example 11.3), so $u_T = \frac{3}{2} f_1$. The *seminormal* vectors u, u_T lie in the orthogonal GZ lines but need not have unit length. In this basis, by Theorem 11.2(ii) with $r = -2$, s_2 acts by

$$\begin{pmatrix} -\frac{1}{2} & \frac{3}{4} \\ 1 & \frac{1}{2} \end{pmatrix},$$

whose first column has length $\sqrt{5}/2$: this seminormal matrix is not orthogonal. The squared norms are $\|u\|^2 = 6$ and $\|u_T\|^2 = 9/2$ (Example 11.3), so normalising gives the basis (v_{T^λ}, v_T), in which s_2 acts by

$$\begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.$$

Fixing lengths does not fix phases: in the orthonormal GZ basis $(v_{T^\lambda}, -v_T)$ both off-diagonal entries are

$$-\frac{\sqrt{3}}{2}.$$

12.4 Worked example

We compute the matrices of S_4 on $V^{(3,1)}$. Its tableaux, content vectors and axial distances are

	T_1 1 2 3 4	T_2 1 2 4 3	T_3 1 3 4 2
$\alpha(T)$	$(0, 1, 2, -1)$	$(0, 1, -1, 2)$	$(0, -1, 1, 2)$
$r,\ s_1$	1	1	-1
$r,\ s_2$	1	-2	2
$r,\ s_3$	-3	3	1

with $T_1 = T^{(3,1)}$, $T_3 = s_2 T_2$ and $T_2 = s_3 T_1$. As

$$\sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}, \qquad \sqrt{1 - \frac{1}{9}} = \frac{2\sqrt{2}}{3},$$

Theorem 12.9 gives, in the basis $(v_{T_1}, v_{T_2}, v_{T_3})$,

$$D(s_1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

$$D(s_2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix},$$

$$D(s_3) = \begin{pmatrix} -\frac{1}{3} & \frac{2\sqrt{2}}{3} & 0 \\ \frac{2\sqrt{2}}{3} & \frac{1}{3} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Branching. $D(s_1)$ and $D(s_2)$ fix v_{T_1}, and on (v_{T_2}, v_{T_3}) they are the matrices of $V^{(2,1)}$, as removing 4 from T_2 and T_3 leaves the two tableaux of shape $(2,1)$:

$$\operatorname{Res}^{S_4}_{S_3} V^{(3,1)} \cong V^{(3)} \oplus V^{(2,1)},$$

as the branching rule of Theorem 12.6 says.

A Coxeter relation. Multiplying out, both $D(s_2)D(s_3)D(s_2)$ and $D(s_3)D(s_2)D(s_3)$ equal

$$\begin{pmatrix} -\frac{1}{3} & -\frac{\sqrt{2}}{3} & \frac{\sqrt{6}}{3} \\ -\frac{\sqrt{2}}{3} & \frac{5}{6} & \frac{\sqrt{3}}{6} \\ \frac{\sqrt{6}}{3} & \frac{\sqrt{3}}{6} & \frac{1}{2} \end{pmatrix},$$

the matrix of $(2\,4)$. Its trace is 1.

The character. So

$$\chi^{(3,1)}\big((1\,2)\big) = \operatorname{tr} D(s_1) = 1,$$

as for $(2\,4)$. In § 6.6's table the irreducibles of degree 3 are W, with $\chi_W((1\,2)) = 1$, and $\operatorname{sgn} \otimes W$, with -1. So $V^{(3,1)} \cong W$, as Lecture 8 promised.

Self-check. Find the matrices of s_1, s_2, s_3 on $V^{(2,2)}$, and $\chi^{(2,2)}((1\,2))$. *Answer:*

	T $\begin{array}{\|c\|c\|}\hline 1&2\\\hline 3&4\\\hline\end{array}$	$T' = s_2T$ $\begin{array}{\|c\|c\|}\hline 1&3\\\hline 2&4\\\hline\end{array}$
$\alpha(T)$	$(0,1,-1,0)$	$(0,-1,1,0)$
$r,\ s_1$	1	-1
$r,\ s_2$	-2	2
$r,\ s_3$	1	-1
$s_1v,\ s_3v$	v_T	$-v_{T'}$

The last row says that each of s_1, s_3 sends v_T to v_T and $v_{T'}$ to $-v_{T'}$; $D(s_2)$ is Example 12.10's orthogonal matrix, since its axial distances are again $-2, 2$. The character value is the trace,

$$\chi^{(2,2)}\big((1\,2)\big) = 1 - 1 = 0,$$

as for $\chi_E \circ \pi$ in § 6.6. The two weights of $\rho_E \circ \pi$ are the displayed content vectors (§ 11.5), so classification identifies $V^{(2,2)} \cong \rho_E \circ \pi$. Also $D(s_1 s_3) = \mathbb{1}$, as $s_1 s_3$ lies in V_4, the kernel of $\rho_E \circ \pi$.

Summary.

- *The main results:* the tableaux of one shape are linked (Theorem 12.5); $\mathrm{Spec}(n) = \mathrm{Cont}(n)$, the irreducibles V^λ and branching for $n \geq 2$ (Theorem 12.6); Young's orthogonal form (Theorem 12.9).
- *Handed on:* the classification, branching and Young's graph to Lectures 13 and 14.
- *Next:* Lecture 13 finds all the characters χ^λ at once.

References

Vershik and Okounkov, *A new approach to the representation theory of the symmetric groups. II* (arXiv:math/0503040): Young's graph, pp. 19–20; Prop. 5.3 and Lemma 5.4, p. 20; Cor. 5.5, p. 21; Thm 5.8, p. 22; §6, pp. 22–23. Ceccherini-Silberstein, Scarabotti and Tolli, *Representation Theory of the Symmetric Groups*: Thm 3.3.7, p. 106; Prop. 3.3.9 and Cor. 3.3.11, p. 107; §3.4.1–3.4.2, pp. 110–113.[1]

[1] They write $\mathrm{Tab}(\lambda)$ for $\mathrm{SYT}(\lambda)$ and S^λ for V^λ, and order the tableaux by the length of σ_T as a word in the s_i (VO p. 22, CST Prop. 3.4.1).